Nate, Thomas, Alvin, and Ronald
Penguin Jacket
GE 1502

**Design Story**

Tasked with creating a biomimetic product, our team decided to approach the challenge by reverse engineering the task. Instead of starting with a problem and searching for a biomimetic solution, we first explored interesting animal adaptations and then brainstormed how those natural features could inspire new products. At first, we were fascinated by the way helicopter seeds spiral through the air. We came up with several ideas for how this mechanism could be applied, but ultimately, none of the use-cases we imagined felt practical or achievable within the scope of our project.

Looking for a new direction, we shifted our focus to the penguin's remarkable temperature regulation system. Penguins can survive in harsh climates by diverting blood flow to specific parts of their bodies, effectively controlling where heat is retained or released. We were inspired by this adaptation and began to imagine a jacket that could mimic this function, using temperature sensors and heating elements to direct warmth to where it's needed most. Deciding which animal function to replicate was the hardest part of the project, but our "backwards" approach made the process a bit more manageable.

Our first prototype was very basic, serving mainly as a rough layout for the jacket. We researched which areas of the human torso are most critical for heat and mapped out where the heating pads should go. At the same time, we started thinking about how to build the circuit and provide enough power for effective heating.

For our second prototype, we developed a complete circuit diagram and figured out how to attach the heating pads to the jacket fabric. We planned to sew the pads in using extra fabric we found at EXP. Although this version didn't have a working heating system yet, we had gathered all the necessary materials and had a clear plan for moving forward.

When we reached our third prototype, we realized that adding an extra layer of fabric between the heating pads and the body was causing too much thermal energy loss, especially since we were already limited by the Arduino's voltage output. To solve this, we decided to glue the heating pads directly onto the jacket, minimizing unnecessary energy loss. While glue wasn't the most sophisticated solution, it worked well given our time and resource constraints. Unfortunately, this prototype revealed a new problem: the heating pads we had ordered didn't work at all. Despite this setback, nearly every other aspect of the jacket was ready.

After ordering new heating pads and confirming they worked with a DC power supply, Alvin and Ronald finished the wiring and installation, and we filmed the demonstration for our project video. In the end, we successfully created a heated jacket inspired by penguin biology. However, we recognize that both the appearance and functionality could be improved. With more time and access to industrial equipment, we would have sewn the wiring into the fabric for a more seamless and durable design. Waterproofing the jacket is another improvement we would have liked to make if time had allowed.

Overall, this project was a great exercise in creative problem-solving and showed us both the potential and the challenges of translating biological systems into real-world engineering solutions.